Rise of the "Intelligent & Connected" Embedded Devices

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Imagine if your "intelligent car" could get information about traffic jams & road accidents, and suggest you an alternate route! Imagine when you go shopping for groceries in the near future, your "intelligent" shopping cart could check the refrigerator in your house, realize that you are out of milk, and recommend that you take a trip into the milk aisle! How about an energy device at home that monitors all your household appliances & makes recommendations to you about your energy usage?

All of this will soon be possible with a new class of "Embedded" devices which are connected to the internet.

What are Embedded devices?

Simply put, an Embedded device is a computer which is dedicated to a specific task. i.e. it is a computer which does not behave like a computer! For example, a microwave oven contains a computer chip, but its end function is to heat up food. Similarly, a camcorder is a computing device which is meant for creating videos. Everything from watches to cars to cameras to traffic lights to home security systems fall into this category of "Embedded" devices.

Embedded devices touch almost all aspects of our daily lives. According to studies, the average person in the US interacts with more than 25 embedded devices in a day. Did you know that many ATM machines actually have an Intel Core2Duo chip inside (the same one that's in your laptop) & run a version of the Windows operating system? Did you know that when you get an X-ray or an ultrasound done at a hospital, you are probably using a device with an Intel Pentium4 chip inside? Most people are not aware of the "intelligence" inside these devices.... As the late Marc Weiser, CTO of Xerox PARC, once observed: "The most profound technologies are those that seem to disappear as they weave themselves into the fabric of our everyday lives".

The future is "connected"

The internet is bringing transformative changes to the embedded world. The era of intelligent connectivity for everyday objects has started & will pick up pace rapidly. According to a recent IDC study, there will be 15 billion devices connected to the internet by 2015, most of which will be embedded devices such as cars, home media phones, digital signs and shopping carts, medical devices, factory robots, intelligent wind turbines, etc.

We know from history that once devices get connected to the internet, a whole host of new usages & functions begin to appear. For example, today a refrigerator or a washing machine is a single function device with specific hard-coded functions. The moment you connect it to the internet, your refrigerator can now be remotely accessed from your workplace & you could set it up to tell you when you have run out of eggs!

Some game-changing embedded products coming in the near future

In Vehicle Infotainment (IVI): Imagine if you could download your favorite movie, access the internet, figure out the fastest route to your destination based on current traffic conditions, have your children play video games - all while you are driving in the car. This will soon be possible with an IVI system in your car powered by Intel's Atom processor. Some of the interesting apps are location based services, state-of-the-art navigation systems, vehicle-to-vehicle communication, multimedia content, etc. Obviously, this is in very early stages in India but we are pretty excited about the long term prospects.

Portable Embedded Devices for Financial Inclusion: A large part of India resides in rural areas, which are not well served by bank branches. In order to allow rural India to participate in the banking system, we will see an explosion of portable embedded banking devices. These are usually ruggedized portable devices with a biometric scanner & integrated printer/card reader, and run the same apps which are run on a PC. A bank employee goes into the village with this device, meets villagers, enrolls them with a bank account & handles transactions. The transactions are synchronized with the bank servers over GPRS.

Retail: Imagine walking around a retail store and coming across a kiosk, which suddenly springs into life with your image! The system goes on to show you various items for you to choose, based on your previous shopping history & the current store inventory. You may be able to "try on" different clothing and match items with each other. This next generation shopping experience is actually possible today on the Intel "proof-of-concept" built on an Intel® Core™ i7 processor and using multi-user, multi-touch holographic display technology.

Intel's role in driving the future of Embedded

All these new "connected" devices will require chips with high-performance to be able to support an operating system & run all these new applications. Driven by breakthroughs in microarchitecture and process technology, the same Intel® architecture that is at the heart of the majority of today's PCs can now deliver intelligence and connectivity to billions of these new intelligent, connected devices.

Intel offers a scalable roadmap of high performance processors ranging from the Xeon processor for high performance embedded devices, all the way down to our new embedded Atom family of processors for low-power devices. In addition, Intel's Embedded Alliance is one of the world's most recognized and trusted ecosystems with a host of partners who provide leading edge embedded devices based on Intel Architecture.

In summary, a new era of connected embedded devices is dawning, and will have a profound effect on the way we live & work. We live in exciting times ! ■